

Practical No – 2 Air and heart

1. Load Library and dataset and print DS

- import pandas as pd
- import numpy as np
- air = pd.read_csv("air_quality.csv",sep=";")
- air.head()
- heart = pd.read_csv("heart.csv")
- heart.head()

2. See Information of Datasets

- heart.info()
- heart.describe()
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- air.describe()

3. Check Number of null values

- heart.isnull().sum()
- air.isnull().sum()

4. Data cleaning

- air.drop_duplicates()
- heart.drop_duplicates()
- print(air.columns) → print columns of DS
- air.drop(columns=['Unnamed: 15','Unnamed: 16'],inplace=True,errors='ignore')
- air.head() → Remove Unnamed columns
- heart.replace(-200, np.nan, inplace=True) → It replaces all values - **200 with NaN (missing value)** in the dataset.
- air.replace(-200, np.nan,inplace=True)
- air = air.fillna(air.mean(numeric_only=True))
- air.head()
- heart = heart.fillna(heart.mean(numeric_only = True))
- heart.head() → Replaces all **NaN (missing values)** with the calculated mean values

5. Data Integration

- merge = pd.concat([air,heart],axis=0,ignore_index=True)
- merge.head(5)

6. Data Transformation

- air['T']=air['T'].str.replace(",",".")

- `air.head()`
- `air['Temp_F']=(air['T'] * 9/5 +32)`
- `air[['Temp_F','T']].head()`
- `heart["age_group"]=pd.cut(heart['age'],bins=[0,30,50,100],labels=["young",'middle','old'])`
- `heart.head()` → Create new column age_group

7. Error correcting

- `heart = heart[(heart['age']>0)&(heart['age']<120)]`
- `heart = heart[(heart['chol'])>0]`
- `heart.head()`
- `air['RH'] = air['RH'].str.replace(',', '.')`
- `air['AH'] = air['AH'].str.replace(',', '.')`
- `air.head()`

8. Data model building

- `heart.corr(numeric_only=True)` → This code calculates the **correlation between all numeric columns** in the heart dataset.
- `heart.groupby('age_group')['chol'].mean()` → This code calculates the **average cholesterol (chol) for each age group** in the heart dataset.
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